

EXPERIMENT 4

Course Title: Software Design Principles (SDP)

Selected Project: Online Mutual Fund Portal

Document: Software Requirements Specification (SRS)

Team Members: 4 Collaborators (Team Project)

Author / Contributor: Krishna Kumar

Aim

To understand the structure, significance, and quality attributes of a Software Requirements Specification (SRS) and prepare a comprehensive SRS document for the **Online Mutual Fund Portal** based on the requirements identified during the previous experiments.

Objectives

1. Understand what a Software Requirements Specification (SRS) is and examine its significance in software development.
 2. Study the key characteristics of a high-quality SRS such as correctness, completeness, consistency, unambiguity, verifiability, modifiability and traceability.
 3. Transform the stakeholder needs, functional requirements and non-functional requirements identified in Experiment 3 into a formal SRS document.
 4. Define external interface requirements including user, hardware, software and communication interfaces.
 5. Specify detailed system features with priorities, stimulus/response sequences and functional requirement identifiers.
 6. Develop analysis models such as Use Case Diagram, Context Level 0 DFD, Level 1 DFD and Entity Relationship Diagram.
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Background & Theoretical Framework

1. What is a Software Requirements Specification (SRS)?

A Software Requirements Specification (SRS) is a formal document that describes the purpose, functionality, behavior, constraints and requirements of a software system.

It acts as a common reference between stakeholders, software developers, testers, project managers and administrators.

For the Online Mutual Fund Portal, the SRS defines requirements related to user registration, mutual fund discovery, fund comparison, investment management, portfolio management, transaction tracking and administrative operations.

2. Significance of SRS

- **Accurately Defining System Requirements:** Clearly documents the functional and non-functional requirements of the portal.
- **Reference Point Across SDLC:** Provides a common reference during design, development, testing and maintenance.
- **Facilitating Team Communication:** Helps investors, administrators, developers and testers understand the expected system behavior.
- **Reducing Development Errors:** Identifies missing or ambiguous requirements before implementation begins.
- **Supporting Testing:** Provides a basis for preparing test cases and validating system functionality.

3. Characteristics of a Good SRS Document

A good SRS should have the following characteristics:

1. **Correctness** – Requirements should represent actual stakeholder needs.
 2. **Completeness** – All major functional and non-functional requirements should be documented.
 3. **Consistency** – Requirements should not contradict each other.
 4. **Unambiguity** – Each requirement should have only one interpretation.
 5. **Verifiability** – Requirements should be testable.
 6. **Modifiability** – Requirements should be organized so that changes can be made easily.
 7. **Traceability** – Requirements should be traceable from stakeholder needs through design and testing.
 8. **Prioritization** – Requirements should be categorized according to importance.
 9. **Understandability** – Requirements should be understandable to technical and non-technical stakeholders.
 10. **Testability** – Requirements should provide a clear basis for software testing.
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Software Requirements Specification (SRS)

For

Online Mutual Fund Portal

Version: 1.0

Prepared by: Krishna Kumar

Organization: KIET Group of Institutions / Department of Information Technology

Project Domain: Financial Technology / Investment Management

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Revision History

Name	Date	Reason for Changes	Version
Krishna Kumar	September 2026	Initial SRS based on Experiments 2 and 3	1.0

1. Introduction

1.1 Purpose

This document specifies the software requirements for Version 1.0 of the **Online Mutual Fund Portal**.

The SRS defines the functional capabilities, non-functional requirements, external interfaces, system constraints, database requirements and analysis models required for developing the portal.

The system provides facilities for investors to browse mutual funds, search and filter schemes, compare funds, invest, manage portfolios and view transaction history.

Administrators can manage mutual fund schemes, user accounts, investment information and transactions.

1.2 Document Conventions

- Functional requirements are identified using REQ-x.y.
- Non-functional requirements are identified using NFR-x.
- Requirements are assigned priorities such as **High, Medium and Low**.
- The word **shall** indicates a mandatory requirement.
- The words **should** and **may** indicate desirable or optional functionality.

1.3 Intended Audience

This document is intended for:

- **Software Developers:** To implement frontend, backend and database functionality.
- **Project Managers:** To monitor development activities and project requirements.
- **Testers:** To prepare test cases and validate system functionality.
- **Administrators:** To understand administrative features.
- **Investors/Users:** To understand available portal functionality.

1.4 Product Scope

The Online Mutual Fund Portal is a web-based investment management system designed to provide users with a centralized platform for mutual fund discovery and investment management.

The system includes:

- Secure user registration and authentication.
- Mutual fund browsing and searching.
- Mutual fund filtering.
- Detailed fund information.

- Fund comparison.
- Mutual fund investment.
- Portfolio management.
- Investment performance tracking.
- Transaction history.
- Administrator management of mutual fund schemes.
- User and transaction management.

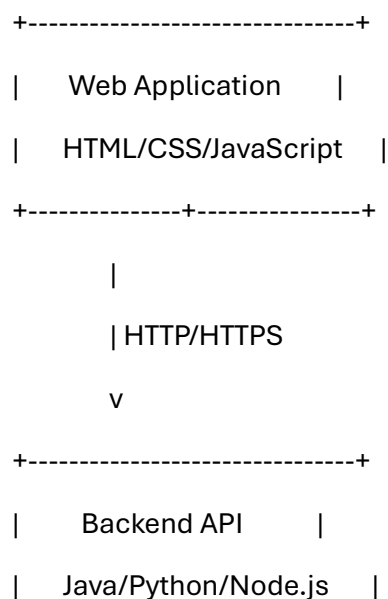
1.5 References

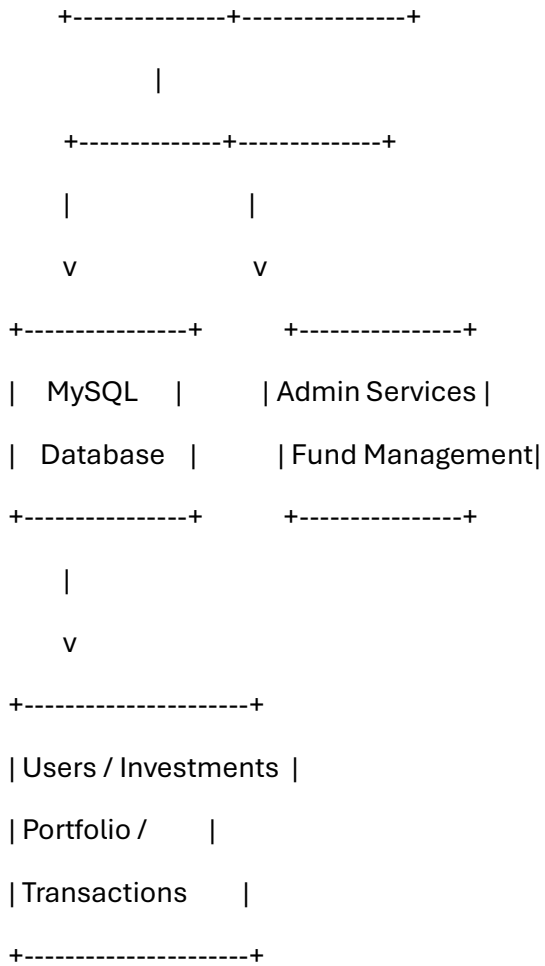
1. IEEE 830 – Recommended Practice for Software Requirements Specifications.
2. Karl E. Wieggers – Software Requirements.
3. Experiment 2 – Problem Identification and Feasibility Analysis.
4. Experiment 3 – Stakeholder Identification and Requirement Gathering Report.
5. Project documentation for Online Mutual Fund Portal.

2. Overall Description

2.1 Product Perspective

The Online Mutual Fund Portal is a web-based application consisting of a frontend, backend services and a relational database.





2.2 Product Functions

The major functions of the system are:

- User registration and authentication.
- Mutual fund browsing.
- Mutual fund search and filtering.
- Fund detail viewing.
- Mutual fund comparison.
- Investment management.
- Portfolio management.
- Investment performance tracking.
- Transaction history.
- Administrator fund management.

- User management.
- Investment and transaction monitoring.

2.3 User Classes and Characteristics

Investors

- General users who want to explore and invest in mutual funds.
- Require simple navigation and understandable fund information.
- Need secure access to their portfolios and transactions.

Beginner Investors

- Users with limited investment experience.
- Require clear fund information and comparison features.
- Need simple interfaces and easy navigation.

Experienced Investors

- Users who manage multiple investments.
- Require portfolio tracking and investment performance information.

Administrators

- Manage system data and mutual fund schemes.
- Manage users and monitor transactions.
- Require administrative dashboards and secure access.

2.4 Operating Environment

- **Frontend:** HTML5, CSS3 and JavaScript.
- **Backend:** Java, Python or Node.js.
- **Database:** MySQL.
- **Client:** Desktop, laptop, tablet and mobile browsers.
- **Operating System:** Windows, Linux, macOS or other systems supporting modern browsers.
- **Development Tools:** VS Code / IntelliJ IDEA.
- **Version Control:** Git and GitHub.

2.5 Design and Implementation Constraints

- User information must be securely stored.
- Authentication must be implemented for protected operations.
- Users must only access their own investment information.
- Administrator functionality must be restricted to authorized users.
- Database operations should maintain data consistency.
- Sensitive configuration information such as passwords and database credentials should not be stored directly in source code.
- The application should follow modular software design principles.

2.6 User Documentation

The system documentation may include:

- User registration and login guide.
- Mutual fund search guide.
- Investment procedure guide.
- Portfolio management guide.
- Transaction history guide.
- Administrator management guide.

2.7 Assumptions and Dependencies

- Users have an internet connection.
- Users have access to a modern web browser.
- Mutual fund information is maintained by authorized administrators or external data providers.
- Investment-related payment processing may depend on an external payment service if integrated.
- Database availability is required for storing user and investment information.

3. External Interface Requirements

3.1 User Interfaces

The system shall provide:

User Portal

- Login/Register page.
- User dashboard.
- Mutual fund listing.
- Search and filter interface.
- Fund details page.
- Fund comparison page.
- Investment page.
- Portfolio dashboard.
- Transaction history page.

Administrator Portal

- Admin login.
- Admin dashboard.
- Mutual fund management.
- User management.
- Investment monitoring.
- Transaction monitoring.

3.2 Hardware Interfaces

The system does not require specialized hardware.

It should support:

- Desktop computers.
- Laptops.
- Tablets.
- Smartphones.
- Standard keyboard and mouse/touch interfaces.

3.3 Software Interfaces

- Web browser.
- Backend application server.
- MySQL database.

- REST API services.
- Git/GitHub for development and version control.

3.4 Communication Interfaces

- The frontend communicates with backend services using HTTP/HTTPS.
 - REST APIs use JSON for data exchange.
 - Secure HTTPS communication should be used for authentication and investment-related operations.
-

4. System Features

4.1 System Feature 1: User Registration & Authentication

Description and Priority

Allows investors and administrators to register/login and access functionality according to their roles.

Priority: High

Stimulus/Response

- **Stimulus:** User enters login credentials.
- **Response:** System validates the credentials.
- **Stimulus:** Valid credentials are submitted.
- **Response:** User is authenticated and redirected to the appropriate dashboard.
- **Stimulus:** Invalid credentials are submitted.
- **Response:** System displays an authentication error.

Functional Requirements

- **REQ-1.1:** The system shall allow users to register.
 - **REQ-1.2:** The system shall authenticate registered users.
 - **REQ-1.3:** The system shall provide administrator authentication.
 - **REQ-1.4:** The system shall enforce role-based access.
 - **REQ-1.5:** The system shall reject invalid login credentials.
-

4.2 System Feature 2: Mutual Fund Search & Browsing

Description and Priority

Allows users to browse, search and filter available mutual fund schemes.

Priority: High

Stimulus/Response

- **Stimulus:** User enters a fund name or search keyword.
- **Response:** System displays matching mutual fund schemes.
- **Stimulus:** User applies filters.
- **Response:** System displays funds matching the selected criteria.

Functional Requirements

- **REQ-2.1:** Users shall be able to browse mutual fund schemes.
 - **REQ-2.2:** Users shall be able to search for mutual funds.
 - **REQ-2.3:** Users shall be able to filter mutual funds.
 - **REQ-2.4:** The system shall display available fund information.
-

4.3 System Feature 3: Fund Details & Comparison

Description and Priority

Allows users to view detailed information and compare different mutual fund schemes.

Priority: High

Stimulus/Response

- **Stimulus:** User selects a mutual fund.
- **Response:** System displays detailed information.
- **Stimulus:** User selects multiple funds for comparison.
- **Response:** System displays the selected funds for comparison.

Functional Requirements

- **REQ-3.1:** The system shall display detailed mutual fund information.
- **REQ-3.2:** Users shall be able to select funds for comparison.
- **REQ-3.3:** The system shall display comparison information for selected funds.
- **REQ-3.4:** The system shall allow users to remove funds from comparison.

4.4 System Feature 4: Investment Management

Description and Priority

Allows authenticated users to invest in selected mutual fund schemes and maintain investment records.

Priority: High

Stimulus/Response

- **Stimulus:** User selects a mutual fund and investment amount.
- **Response:** System validates the investment information.
- **Stimulus:** Investment is successfully processed.
- **Response:** System creates an investment record.
- **Stimulus:** User views investments.
- **Response:** System displays the user's investment information.

Functional Requirements

- **REQ-4.1:** Authenticated users shall be able to initiate investments.
- **REQ-4.2:** The system shall validate investment information.
- **REQ-4.3:** The system shall maintain investment records.
- **REQ-4.4:** The system shall associate investments with the correct user account.
- **REQ-4.5:** Investment status shall be maintained in the system.

4.5 System Feature 5: Portfolio & Performance Tracking

Description and Priority

Allows users to view their investments and track portfolio performance.

Priority: High

Stimulus/Response

- **Stimulus:** User opens the portfolio.
- **Response:** System retrieves the user's investment records.
- **Stimulus:** User requests performance information.

- **Response:** System displays available portfolio performance information.

Functional Requirements

- **REQ-5.1:** Users shall be able to view their current investments.
 - **REQ-5.2:** The system shall display portfolio information.
 - **REQ-5.3:** Users shall be able to view investment history.
 - **REQ-5.4:** The system shall display available investment performance information.
-

4.6 System Feature 6: Transaction History

Description and Priority

Provides users with records of their previous investment-related transactions.

Priority: Medium

Stimulus/Response

- **Stimulus:** User opens transaction history.
- **Response:** System retrieves transactions associated with the authenticated user.
- **Stimulus:** User selects a transaction.
- **Response:** System displays relevant transaction details.

Functional Requirements

- **REQ-6.1:** The system shall record investment transactions.
 - **REQ-6.2:** Users shall be able to view their transaction history.
 - **REQ-6.3:** Transaction information shall include relevant transaction details.
 - **REQ-6.4:** Users shall only be able to access their permitted transaction information.
-

4.7 System Feature 7: Administrator Management

Description and Priority

Allows administrators to manage mutual fund schemes, user accounts and system transactions.

Priority: High

Stimulus/Response

- **Stimulus:** Administrator logs into the admin portal.
- **Response:** System authenticates the administrator.
- **Stimulus:** Administrator adds a new fund.
- **Response:** System validates and stores the fund information.
- **Stimulus:** Administrator views transactions.
- **Response:** System displays authorized transaction information.

Functional Requirements

- **REQ-7.1:** Administrators shall be able to add mutual fund schemes.
 - **REQ-7.2:** Administrators shall be able to update fund information.
 - **REQ-7.3:** Administrators shall be able to remove mutual fund schemes.
 - **REQ-7.4:** Administrators shall be able to manage user accounts.
 - **REQ-7.5:** Administrators shall be able to monitor investments.
 - **REQ-7.6:** Administrators shall be able to monitor transactions.
-

5. Other Non-Functional Requirements

5.1 Performance Requirements

- **NFR-1:** Common operations such as login, fund search and portfolio viewing should provide a reasonably fast response.
- **NFR-2:** Database queries should be optimized to support increasing numbers of users and funds.
- **NFR-3:** The system should support multiple users accessing the portal simultaneously.

5.2 Security Requirements

- **NFR-4:** Authentication shall be required for protected operations.
- **NFR-5:** Role-based authorization shall restrict administrator functionality.
- **NFR-6:** Users shall only access their own portfolio and transaction information.
- **NFR-7:** Passwords shall be stored using secure password-hashing mechanisms.

- **NFR-8:** Communication involving sensitive information should use HTTPS.
- **NFR-9:** Database credentials and other sensitive configuration values should be protected using environment variables.

5.3 Reliability Requirements

- **NFR-10:** Investment records shall be stored consistently.
- **NFR-11:** Transaction information shall not be lost during normal system operation.
- **NFR-12:** Database operations should maintain data integrity.

5.4 Software Quality Attributes

- **Usability:** The portal should provide simple navigation.
- **Maintainability:** The application should use modular components.
- **Scalability:** The system should support increasing numbers of users and schemes.
- **Reliability:** Investment and transaction information should remain consistent.
- **Security:** Authentication and authorization mechanisms should protect user information.
- **Portability:** The web application should work across modern browsers.

5.5 Business Rules

- **BR-1:** Only registered users can access protected investment functionality.
- **BR-2:** Users can view only their own portfolio and transaction records.
- **BR-3:** Only authorized administrators can modify mutual fund scheme information.
- **BR-4:** Investment records must be associated with the correct user account.
- **BR-5:** Transaction records should be maintained for investment tracking.

6. Other Requirements

6.1 Database Requirements

The MySQL database shall store information related to:

- Users.

- Administrators.
- Mutual fund schemes.
- Investments.
- Portfolios.
- Transactions.

The database should use:

- Primary keys.
- Foreign keys.
- Appropriate indexes.
- Data validation.
- Referential integrity.

Proposed Main Tables

Users

|

|----< Investments

|

|----< Transactions

|

MutualFunds -+

|

|----< Investments

6.2 Future Integration Requirements

Future versions of the system may include:

- Online payment gateway integration.
- External mutual fund data APIs.
- SIP management.
- Email/SMS notifications.
- Advanced portfolio analytics.

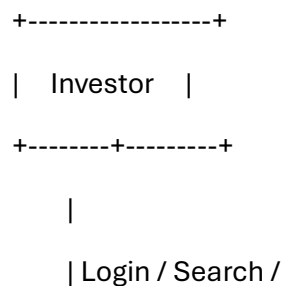
- Mobile application.
- Additional investment products.

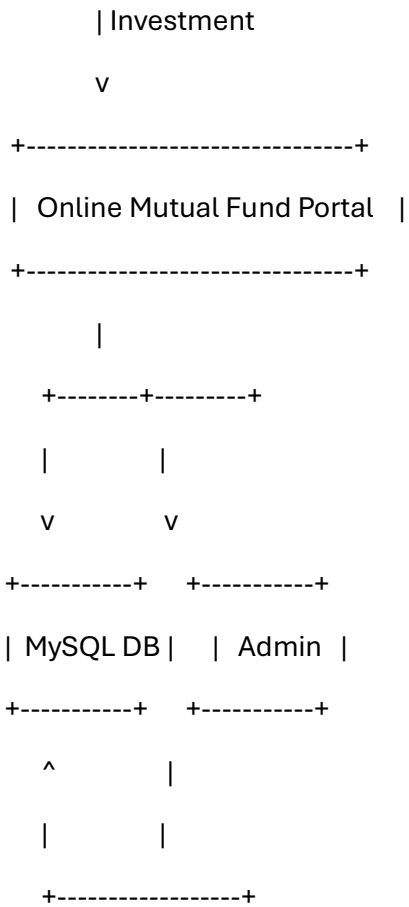
Appendix A: Glossary

Term	Definition
API	Application Programming Interface
Admin	Administrator who manages the portal
Authentication	Process of verifying a user's identity
Authorization	Process of determining permitted actions
Fund	Mutual fund investment scheme
Investment	Amount invested by a user in a mutual fund
Portfolio	Collection of a user's investments
SRS	Software Requirements Specification
SQL	Structured Query Language
UI	User Interface
RBAC	Role-Based Access Control
NFR	Non-Functional Requirement
REQ	Functional Requirement Identifier

Appendix B: Analysis Models

Model 1: Context Level 0 DFD





Model 2: Level 1 DFD

Investor



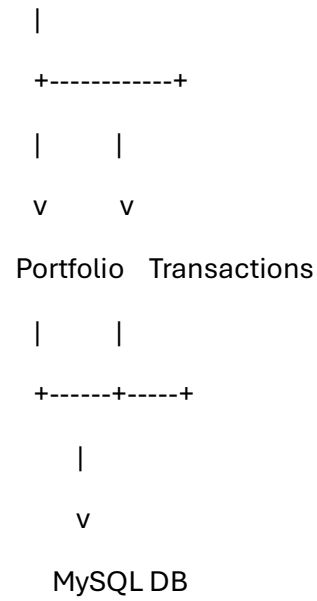
Authentication Fund Management



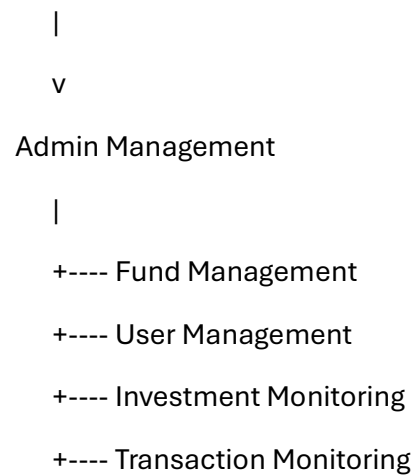
User Database Mutual Fund DB



Investment Management



Administrator



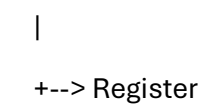
Model 3: Use Case Diagram

Actors

- Investor
- Administrator

Investor Use Cases

Investor



- +--> Login
- +--> Browse Funds
- +--> Search Funds
- +--> Filter Funds
- +--> View Fund Details
- +--> Compare Funds
- +--> Invest
- +--> View Portfolio
- +--> Track Performance
- +--> View Transactions

Administrator Use Cases

Administrator

- |
- +--> Login
- +--> Add Fund
- +--> Update Fund
- +--> Remove Fund
- +--> Manage Users
- +--> Monitor Investments
- +--> Monitor Transactions

Model 4: Entity Relationship Diagram

+-----+

| USERS |

+-----+

| user_id (PK) |

| name |

| email |

| password |

+-----+-----+

|

| 1

|

| M

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| INVESTMENTS |

+-----+

| investment_id |

| user_id (FK) |

| fund_id (FK) |

| amount |

| date |

| status |

+-----+-----+

|

| M

|

| 1

+-----v-----+

| MUTUAL_FUNDS |

+-----+

| fund_id (PK) |

| fund_name |

| category |

| nav |

| description |

+-----+

```

+-----+
| TRANSACTIONS |
+-----+
| transaction_id |
| user_id (FK) |
| investment_id |
| amount      |
| date        |
| type        |
| status      |
+-----+

```

Appendix C: To Be Determined (TBD) List

Item	Section	Description	Resolution Phase
TBD-1	Investment Management	Exact payment gateway to be integrated	Implementation Phase
TBD-2	Fund Information	External source/API for live mutual fund information	Design Phase
TBD-3	Notifications	SMS/email provider for investment notifications	Implementation Phase
TBD-4	Portfolio	Detailed methodology for calculating performance	Design Phase

Observations & Analytical Insights

1. Preparing the SRS converts stakeholder requirements identified in earlier experiments into formal software requirements.
2. Functional requirement identifiers such as REQ-1.1 make requirements easier to track and test.

3. DFD, Use Case and ER models help clarify system boundaries, data flow and relationships between entities.
4. Defining non-functional requirements establishes important quality attributes such as security, reliability, usability, performance and scalability.
5. The SRS provides a common technical reference for developers, testers, administrators and other stakeholders.

Result

The **Software Requirements Specification (SRS) for the Online Mutual Fund Portal** was successfully formulated. The document establishes the system's functional requirements, non-functional requirements, external interfaces, system features, database requirements, business rules and analysis models, providing a structured foundation for the design, development and testing of the Online Mutual Fund Portal.

Team Contribution Note

- **Course:** Software Design Principles (SDP)
- **Project:** Online Mutual Fund Portal
- **Team:** 4 Members
- **Members:**
 1. Krishna Kumar
 2. Rahul Kumar Yadav
 3. Pratik Singh
 4. Rishav Yadav